Sports Science Research Update

March 2023

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Pajamas, polysomnography and professional athletes: The role of sleep tracking technology in sport. Sports, 11(14): 1-17, 2023.

Regular-season injury rates in the National Football League after an attenuated preseason due to COVID-19. The Orthopaedic Journal of Sports Medicine, 10(11): 1-4, 2022.

Relationships between repeated sprint ability, countermovement jump and thermography in elite football players. Sensors, 23(632): 1-9, 2023.

Scheduling concurrent training 48 versus 72 hours after simulated match play: Effects on neuromuscular function and fatigue. Medicine and Science in Sports and Exercise, 55(2): 301-310, 2023.

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The use of biologics in NFL athletes: An expert consensus of NFL team physicians. The Orthopaedic Journal of Sports Medicine, 11(2): 1-15, 2023.

Two or four weeks acute:chronic workload ratio is more useful to prevent injuries in soccer? Applied Sciences, 13(495): 1-8, 2023.

Weekly external load performance effects on sports injuries of male professional football players. International Journal of Environmental Research and Public Health, 20(1121): 1-12, 2023.

Sports Science Research Update

A commentary of factors related to player availability and its influence on performance in elite team sports. Frontiers in Sports and Active Living, Jan. 16: 1-18, 2023.

Overview - Elite performance and sporting success are often the result of optimal integration and synergy of all components of sports preparedness (i.e., health, technical and tactical skills, bioenergetic and neuromuscular abilities and capacities, anthropometric characteristics, cognition, emotions, creativity, or personality), which evolve because of systematic long-term sports preparation. However, the relative importance of these characteristics varies between individual and team sports. While some individual sports require a high standard of bioenergetic and neuromuscular abilities and capacities, team sports performance is closely related to technical and tactical skills, which may compensate for weakness within the fitness level. Nonetheless, successful team sport performances seem to be much more dependent on the interaction among a wide range of factors than on the maximum development of one or two factors in isolation. In team sports, elite performance emerges from the interaction among the individual parts to overcome the opponent during competition. Sports may be categorized according to the degree of predictability of the environment that they are played in. Team sports occur in highly unpredictable environments due to the interactions with both teammates and opponents, with performance dealing with this unpredictability. Thus, it is important to have a clear understanding of the integrative systems and the principles that rule their interactions with the environment, keeping in mind the main aim of the process: developing the diversity/unpredictability potential of athletes/ teams to afford the emergence of rich patterns of behavior from players to adapt quickly and effectively in dynamically changing and unpredictable environments. Performance in team sports is affected by several factors that affect the organization of training and competitions. These include, for example, COVID-19 cases, PCR tests, air flights and their effects prior to competition, injuries, or match-congested schedules. The interaction among these factors may also influence player availability. The concept of player availability is a common one in elite team sports. Available players can be considered the ones who are injury-free and ready to compete whether the head coach chooses to put them on the lineup. Thus, an available state would be when a player is fit and recovered enough to compete. On the other hand, player unavailability would be considered a state which includes injury, sanction or suspension, or other reasons that would keep a player out of match. However, this topic needs to be explored more in elite team sport environments. Considering previous enriching work, it remains important to further progress and provide academic knowledge in order to support coaches/managers, strength and conditioning coaches, sport scientists, and medical team members (e.g., doctors, physicians, and physiotherapists) in their working environments. While widely-advocated scientific groundwork is considered throughout this manuscript, the main aim of this opinion article is to provide a review of factors related to player availability and its influence on performance in elite team sports. Finally, some

practical suggestions and recommendations are provided to deal with constant alterations in player's availability and performance fluctuations.

Sports Science Research Update

Acute adductor muscle injury: A systematic review on diagnostic imaging, treatment, and prevention. The American Journal of Sports Medicine, Ahead of Print: 1-13, 2023.

Background: Controversies remain regarding the diagnosis, imaging, and treatment of acute adductor injuries in athletes.

Purpose: To investigate the diagnostic imaging, treatment, and prevention of acute adductor injuries based on the most recent and relevant scientific evidence.

Study Design: Systematic review; Level of evidence, 4.

Methods: The PubMed and Web of Science databases were searched according to the Preferred Reporting Items for Systematic Reviews and Meta-Analyses guidelines to identify articles studying acute adductor injury in athletes. Inclusion criteria were original publication on acute adductor injury in amateur or professional athletes, level 1 to 4 evidence, mean patient age .15 years, and results presented as return-to-sport, pain, or functional outcomes. Quality assessment was performed with the CONSORT (Consolidated Standards of Reporting Trials) statement or the methodological index for non-randomized studies criteria. Articles were grouped as imaging, treatment, prevention focused, or mixed.

Results: A total of 30 studies published between 2001 and 2021 were selected, involving 594 male patients with a mean age 26.2 years (range, 16-68 years). The most frequent sports were soccer (62%), basketball (14%), futsal (6%), American football (3%), and ice hockey and handball (2%). Risk factors for acute adductor injury were previous acute groin injury, adductor weakness compared with the uninjured side, any injury in the previous season, and reduced rotational hip range of motion. The frequency of complete adductor muscle tears on magnetic resonance imaging was 21% to 25%. For complete adductor tears, the average time to return to play was 8.9 weeks in patients treated nonoperatively and 14.2 weeks for patients treated surgically. Greater stump retraction was observed in individuals treated surgically. Partial acute adductor tears were treated nonoperatively with physical therapy in all studies in the present systematic review. The average time to return to play was 1 to 6.9 weeks depending on the injury grade. The efficacy of adductor strengthening on preventing acute adductor tears has controversial results in the literature.

Conclusion: Athletes with partial adductor injuries returned to play 1 to 7 weeks after injury with physical therapy treatment. Non-operative or surgical treatment is an acceptable option for complete adductor longus tendon tear.

Sports Science Research Update

Athletes are not at greater risk of death by suicide: A review. Scandinavian Journal of Medicine and Science in Sports, Ahead of Print: 1-17, 2023.

Introduction: Suicide represents a major mental and public health issue. Elite athletes share certain individual and environmental characteristics that may increase their risk for mental illnesses, ultimately leading to suicide. This notion conflicts with the general perception of athletes, being the healthiest representatives of society.

Methods: A comprehensive literature search was carried out through PubMed and Embase databases for relevant publications.

Results: Recent calls for investigating suicidality among athletes resulted in a considerable amount of literature providing some evidence regarding lower rates of suicide among professional and high-performance athletes as well as similar incidence and prevalence of mental conditions, which are known as risk factors for suicide. Nevertheless, special attention is required in this population as pre disposing and precipitating factors might differ from classical features of suicidality in the general population. Sports physicians, sports psychiatrists, and other mental health professionals in elite sports should be aware of early signs of affective disorders, risk of recreational drug abuse, misuse of performance-enhancing medications, sport-specific environmental stressors, serious physical injuries, and presence of physical or mental illness, all of which may increase suicidality. Traumatic brain injury (TBI) is with suicide with higher severity correlated with increased risk. Compared to active athletes, former athletes may have higher rates of suicide due to common life stressors occurring after sports retirement.

Conclusions: The findings suggest a multidisciplinary approach to suicidality in elite athletes, the main goal of which should be the reduction of suicide-related morbidity and mortality. Further research is required to clarify the existing gaps in the current knowledge of the issue. While having lower rates of suicide, athletes share some similar (affective disorders, drug abuse, mental and physical illness) and unique factors (misuse of performance-enhancing substances, sports-related stressors, sports injuries, TBI) putting them at risk of suicide during active career and retirement.

Sports Science Research Update

COVID-19 NFL Injury Prevalence Analysis, A Follow-Up Study. medRxiv, Jan. 12: 1-15, 2023.

Background: In 2020, COVID-19 spread across the world and brought the world to a halt, causing the shutdown of nearly everything in order to prevent its spread. The NFL experienced the same effects of the shutdowns leaving athletes unable to train in some of the most advanced facilities with many of the best trainers in the world. Through a previous study, COVID-19 Return to Sport Injury Prevalence Analysis, it was determined that there was increased injury prevalence during the 2020 season likely due to decreased physiological adaptations within athletes' bodies that resulted from facility shutdowns. Understanding injury epidemiology is vital in the prevention of injuries and the development of return-to-play protocols.

Objective: The objective of this study is to perform a follow up study to COVID-19 Return to Sport Injury Prevalence Analysis in order to examine the longitudinal effects of the COVID-19 pandemic on injury epidemiology. This study will examine if there was a recovery to baseline or lingering effects from the COVID-19 pandemic-induced spike in injuries.

Methods: Injury tallies collected from the 17-week-long 2020 NFL regular season, played after COVID-19 restrictions, were compared with the injury tallies collected from the 18-week-long NFL regular seasons (2021, 2022), in order to determine if there was a change in injury prevalence. An unpaired t-test was conducted to compare the mean injuries per team per week between each of the 2020, 2021, and 2022 regular seasons.

Results: The 2022 and 2021 NFL regular seasons produced lower numbers of total injuries than the 2020 NFL regular season that was impacted by COVID-19. The comparison of the mean number of injuries per team per week of the 2020 season compared with the 2021 regular season was statistically significant (P=.03). The comparison of the 2020 and 2022 regular seasons was also statistically significant (P=.02).

Conclusions: The results of this follow-up study and our previous study show that extended training interruptions have the ability to induce detraining and lead to increased injuries. Additionally, the results of this study show that retraining can occur and lead to injury protective factors. This is the first large scale opportunity to demonstrate the effects of these principles and how they are important to understanding injury epidemiology.

Sports Science Research Update

Do repeated sprints affect the biceps femoris long head architecture in football players with and without an injury history? A retrospective study. Biology, 12(96): 1-11, 2023.

Simple Summary: Football is a sport which imposes an intermittent type of physical exertion, and one of the main physical demands of the game is sprinting, covering mostly short distances on repeated occasions. Sprinting imposes maximum demands, especially on the hamstrings muscle group; this activity frequently contributes to the occurrence of injury to the biceps femoris long head muscle, particularly in the later stages of the game, suggesting that fatigue may play an important role. After the injury, sprint performance and strength might be reduced, and muscle morphology can be altered even after the player returns to competition. Therefore, the aim of the study was to verify the effect of a fatiguing task on sprint performance, the biceps femoris long head muscle morphology and the hamstring muscle force capacity. Moreover, a comparison was performed between injured players and healthy controls and injured versus contralateral limbs. Muscle morphology was assessed through ultrasound techniques, while muscle force was estimated using a dynamometry assessment. It was concluded that injured players tend to be faster than the healthy control group and display shorter biceps femoris long head fascicle length when compared with the contralateral limb and healthy controls.

Abstract: The aim of this study was to compare the biceps femoris long head (BFlh) architecture between football players with (twelve) and without (twenty) history of BFlh injury before and after a repeated sprint task. Fascicle length (FL), pennation angle (PA) and muscle thickness (MT) were assessed at rest and in the active condition before and after the repeated sprint protocol. Athletes with previous BFlh injury showed shorter FL at rest ($p = 0.014$; $\eta 2 p = 0.196$) and active state ($p < 0.001$; $\eta 2 p = 0.413$), and greater PA at rest ($p = 0.002$; $\eta 2 p = 0.307$) and active state ($p < 0.001$; $\eta 2 p = 0.368$) before and after the task. Intra-individual comparisons showed that injured limbs have shorter FL at rest ($p = 0.012$; $\eta 2 p = 0.519$) and in the active state ($p = 0.039$; $\eta 2 p = 0.332$), and greater PA in passive ($p < 0.001$; $\eta 2 p = 0.732$) and active conditions ($p = 0.018$; $\eta 2 p = 0.412$), when compared with contralateral limbs. Injured players, at rest and in the active condition, display shorter BFlh FL and greater PA than contralateral and healthy controls after repeated sprints. Moreover, the BFlh of injured players presented a different architectural response to the protocol compared with the healthy controls.

Sports Science Research Update

Does playing surface affect the risk of anterior cruciate ligament injuries in the National Football League? A look at the 2017-2021 NFL seasons. The Physician and Sportsmedicine, Ahead of Print: 1-11, 2023.

Background: Anterior cruciate ligament (ACL) injuries are devastating injuries for athletes. Prior studies have shown increased ACL injury rates on non-natural surfaces versus natural grass in several sports. The purpose of this study is to calculate the prevalence of ACL injuries in the NFL on natural versus non-natural surfaces to determine if there is a significant increase on non-natural surfaces.

Methods: Accessing publicly available data for NFL seasons beginning with the 2017-2018 season through 2021- 2022 seasons, all ACL injuries with publicly available data concerning timing and playing surface were recorded and categorized according to playing surface. Practice injuries or those without an identifiable playing surface were excluded. Incidence rates, defined as ACL ruptures per game, were calculated. ACL injuries were recorded for each playing surface, as well as the combined category of non-natural grass surface. Odds ratio was calculated to compare the risk of ACL rupture on non-natural surfaces vs natural grass.

Results: During the 2017-2021 NFL seasons, 173 ACL ruptures were identified with known surfaces. Injury rate for non-natural surfaces was 0.134 compared to 0.097 for grass. Injury rate ratio for non-natural vs natural grass surfaces was 1.211, a 21.1% increased risk of ACL injury in the NFL on non-natural surfaces vs natural grass. OR for non-natural surfaces 1.239 (95% CI 0.900-1.704). Based on these findings there is a trend toward increased risk of ACL injury on non-natural grass surfaces, however this did not reach statistical significance.

Conclusion: Numerous published studies show trends towards increasing rates of ACL injuries on non-natural playing surfaces vs natural grass. Based on our findings the difference is not statistically significant, however it does trend toward increased risk of ACL injury with non-natural surfaces. Further studies should be performed with larger sample sizes in order to further determine the risk of non-natural surfaces.

Sports Science Research Update

Effectiveness of conservative interventions after acute hamstrings injuries in athletes: A living systematic review. Sports Medicine, Ahead of Print: 1-21, 2023.

Background: Hamstrings injuries are common in sports and the reinjury risk is high. Despite the extensive literature on hamstrings injuries, the effectiveness of the different conservative (i.e., non-surgical) interventions (i.e., modalities and doses) for the rehabilitation of athletes with acute hamstrings injuries is unclear.

Objective: We aimed to compare the effects of different conservative interventions in time to return to sport (TRTS) and/or time to return to full training (TRFT) and reinjury-related outcomes after acute hamstrings injuries in athletes.

Data Sources: We searched CINAHL, Cochrane Library, EMBASE, PubMed, Scopus, SPORTDiscus, and Web of Science databases up to 1 January, 2022, complemented with manual searches, prospective citation tracking, and consultation of external experts.

Eligibility Criteria: The eligibility criteria were multi-arm studies (randomized and non-randomized) that compared conservative treatments of acute hamstrings injuries in athletes.

Data Analysis: We summarized the characteristics of included studies and conservative interventions and analyzed data for main outcomes (TRTS, TRFT, and rate of reinjuries). The risk of bias was judged using the Cochrane tools. Quality and completeness of reporting of therapeutic exercise programs were appraised with the i-CONTENT tool and the certainty of evidence was judged using the GRADE framework. TRTS and TRFT were analyzed using mean differences and the risk of reinjury with relative risks.

Results: Fourteen studies (12 randomized and two non-randomized) comprising 730 athletes (mostly men with ages between 14 and 49 years) from different sports were included. Nine randomized studies were judged at high risk and three at low risk of bias, and the two non-randomized studies were judged at critical risk of bias. Seven randomized studies compared exercise-based interventions (e.g., L-protocol vs C-protocol), one randomized study compared the use of low-level laser therapy, and three randomized and two non-randomized studies compared injections of platelet-rich plasma to placebo or no injection. These low-level laser therapy and platelet-rich plasma studies complemented their interventions with an exercise program. Only three studies were judged at low overall risk of ineffectiveness (i-CONTENT). No single intervention or combination of interventions proved superior in achieving a faster TRTS/TRFT or reducing the risk of reinjury. Only eccentric lengthening exercises showed limited evidence in allowing a shorter TRFT. The platelet-rich plasma treatment did not

consistently reduce the TRFT or have any effect on the risk of new hamstrings injuries. The certainty of evidence was very low for all outcomes and comparisons.

Conclusions: Available evidence precludes the prioritization of a particular exercise-based intervention for athletes with acute hamstrings injuries, as different exercise-based interventions showed comparable effects on TRTS/TRFT and the risk of reinjuries. Available evidence also does not support the use of platelet-rich plasma or low-level laser therapy in clinical practice. The currently available literature is limited because of the risk of bias, risk of ineffectiveness of exercise protocols (as assessed with the i-CONTENT), and the lack of comparability across existing studies.

Key Points

- As exercise-based interventions showed comparable time to return to full training or matches and the risk of reinjuries, no specific strategy needs to be prioritized when rehabilitating athletes with acute hamstrings injuries. Only eccentric lengthening exercises showed limited evidence in allowing a shorter time to return to full training.
- Platelet-rich plasma injections did not consistently reduce the time to return to full training or have any effect on the risk of new hamstrings injuries. Therefore, platelet-rich plasma has no current value in clinical practice.
- The currently available literature is still limited owing to a risk of bias, poor description, the risk of ineffectiveness of exercise protocols, and a lack of comparability across existing studies. Further studies are clearly warranted to allow stronger conclusions

Sports Science Research Update

Effects of contact/collision sport history on gait in early to mid-adulthood. Journal of Sport and Health Science, Ahead of Print: 1-8, 2022.

Background: To determine the effect of contact/collision sport participation on measures of single-task (ST) and dual-task (DT) gait among early- to middle-aged adults.

Methods: The study recruited 113 adults (34.9 ± 11.8 years, 53.0% female) representing 4 groups. Groups included: (a) former non-contact/collision athletes and non-athletes who are not physically active (n = 28); (b) former non-contact/collision athletes who are physically active (n = 29); (c) former contact/collision sport athletes who participated in high-risk sports and are physically active (n = 29); and (d) former rugby players with prolonged repetitive head impact exposure history who are physically active (n = 27). Gait parameters were collected using inertial measurement units during ST and DT gait. DT cost was calculated for all gait parameters (double support, gait speed, stride length). Groups were compared first using a one-way analysis of covariance. Then a multiple regression was performed for participants in the high-risk sport athletes and repetitive head impact exposure athletes groups only to predict gait outcomes from contact/collision sport career duration.

Results: There were no significant differences between groups on any ST, DT, or DT cost outcomes ($p > 0.05$). Contact/collision sport duration did not predict any ST, DT, or DT cost gait outcomes.

Conclusion: Years and history of contact/collision sport participation does not appear to negatively affect or predict neurobehavioral function in early- to mid-adulthood among physically active individuals.

Sports Science Research Update

Fatigue-induced changes in hamstrings' active muscle stiffness: Effect of contraction type and implications for strain injuries. European Journal of Applied Physiology, Ahead of Print: 1-14, 2022.

Purpose: Hamstring strain injuries may occur due to differential fatigue and compromised mechanical properties among the hamstring muscles. We examined (1) the effect of fatigue on hamstrings active muscle stiffness, and (2) whether contraction type affects active muscle stiffness changes during a submaximal fatiguing task.

Methods: Nine healthy males completed 99 submaximal knee flexions in isometric (ISO), concentric (CON), and eccentric (ECC) conditions. We measured the knee flexor maximal voluntary torque (MVT) (pre/post), shear wave velocity (SWV) during contraction and transverse relaxation times (T2) (pre/post) in biceps femoris long head (BFlh), semitendinosus (ST), and semimembranosus (SM) muscles.

Results: MVT decreased substantially after all conditions (− 18.4 to − 33.6%). The average relative torque sustained during the task was lower in CON than ISO and ECC, but absolute torque was similar. SWV interindividual responses were highly variable across muscles and contraction types. On average, BFlh SWV tended to increase in ISO (0.4 m/s, 4.5%, $p=0.064$) but decreased in ECC condition (− 0.8 m/s, − 7.7%, $p<0.01$). ST SWV decreased in CON (− 1.1 m/s, − 9.0%, $p<0.01$), while it remained unchanged in ISO and ECC. SM SWV decreased in CON (− 0.8 m/s, − 8.1%, $p<0.01$), but it was unaffected in ISO and variable in ECC.

Conclusion: Fatigue has a differential effect on the mechanical properties of the constituent hamstring muscles, as measured with shear wave elastography, depending upon contraction type. We found preliminary evidence that BFlh is more fatigued than ST or SM during eccentric contractions, which may explain its susceptibility to strain injuries.

Sports Science Research Update

Increased incidence of acute Achilles tendon ruptures in the peri-pandemic COVID era with parallels to the 2021-22 NFL season. Knee Surgery, Sports Traumatology, Arthroscopy, Jan. 16: 1-6, 2023.

Purpose: Acute Achilles tendon ruptures (AATRs) are a common sporting injury, whether for recreational athletes or elite athletes. Prior research has shown returning to physical activity after extended periods of inactivity leads to increased rates of musculoskeletal injuries. The purpose of this study was to investigate rates of acute Achilles' tendon ruptures at a single academic institute in the peri-COVID era, with corollary to the recent NFL season.

Methods: A retrospective search was conducted using current procedural terminology to identify the total number of Achilles acute primary repair surgeries performed from years 2017 to 2021. Non-operatively managed AATRs were identified from the same electronic medical record using ICD-10 codes. NFL data were obtained from publicly available sites according to previously validated studies.

Results: A total of 588 patients who sustained AATRs and underwent primary surgical repair were identified, primarily men (75.7%, n=445), with an average age of 43.22±14.4 years. The number and corresponding incidence of AATR repairs per year was: 2017: n=124 (21.1%), 2018: n=110 (18.7%), 2019: n=130 (22.1%), 2020: n=86 (14.6%), 2021: n=138 (23.5%), indicating a 7.5% decrease in rate of AATRs from 2019 to 2020, followed by an 8.9% increase in incidence from 2020 to 2021. Within the NFL, the number of AATRs resulting in an injured reserve stint increased every regular season from 2019 to 2020: n=11 (21.2%), to 2020–2021: n=17 (32.7%), to this past 2021–2022 season: n=24 (46.2%).

Conclusion: AATR surgeries seem to have increased in 2021 following a 2020 COVID pandemic-induced quarantine for recreational athletes at a single academic institution and for professional athletes in the NFL, although these results are of questionable clinical significance. This provides prognostic information when counseling patients and athletes on return to activity or sport.

Level of evidence: Level IV.

Sports Science Research Update

Increased snap counts were not seen prior to Lisfranc injuries in the National Football League. Cureus, 14(12): e32736: 1-7, 2022.

Introduction: Athletes in the National Football League (NFL) subject their bodies to a great deal of physical strain, which places them at an increased risk for injury. The purpose of this study was to determine if there was an increase in snap counts played during a player's injury game or season that may have contributed to an increased risk of sustaining a Lisfranc injury in the NFL. We hypothesized that players who play more snaps than they have in seasons prior will be at an increased risk of injury.

Methods: NFL players with Lisfranc injuries were identified by cross-referencing multiple online resources. Information on a player's position, draft year, draft round selection, height, weight, snap counts, approximate value (AV), quarter of injury (first, second, third, fourth), quarter of a season (games one through four, five through eight, nine through twelve, thirteen through sixteen), and injured foot laterality was collected. A control group of players without a Lisfranc injury was then selected to compare performance data with our injured cohort that returned to play.

Results: Twenty-one NFL players, who met inclusion criteria, sustained a Lisfranc injury between the years 2013 and 2021. Players played significantly fewer snaps before sustaining a Lisfranc injury compared to their season average (33.9 ± 21.9 vs. 50.3 ± 15.8; p=<0.001), but when comparing the number of snaps played per game in their injury year with the number of snaps played per game over their career before injury, there was no significant difference (50.3 ± 15.8 vs. 45.7 ± 17.1; p=0.20). Most injuries occurred in either the first (42.9%) or second (33.3%) quarter of a regular season (games one through eight). During a game, the timing of most injuries was either the second (38.1%) or fourth (33.3%) quarter. There was no significant difference between injured players and controls post-injury between the number of average seasons played, AV, and snaps played per game.

Conclusion: Increased snap counts were not seen prior to Lisfranc injuries in the NFL.

Sports Science Research Update

Neuromechanics of the rate of force development. Exercise and Sport Sciences Reviews, 51(1): 34-42, 2023.

Abstract - The rate at which an individual can develop force during rapid voluntary contractions can be influenced by both the neural drive to a muscle and its intrinsic musculotendinous properties. We hypothesize that the maximal rate of force development across human individuals is mainly attributable to the rate of motor unit recruitment.

Key Points

• The rate of force development of human individuals varies greatly across contractions and muscles.

• Here we hypothesize that the rate of motor unit recruitment dictates the inter-subject variability in generating force rapidly.

• Moreover, we discuss evidence that the neural drive to the muscle and the muscle components during rapid contractions behave in a parallel. The tight link between neural and muscular properties determines the association between neural drive and rate of force development during the early and late phases of force generation.

• We discuss the current evidence from computer simulations and experimental recordings of voluntary force levels and motor unit discharge characteristics during rapid contractions.

Sports Science Research Update

Nutritional modulation of sleep latency, duration, and efficiency: A randomized repeated-measures, double-blind deception study. Medicine and Science in Sports and Exercise, 55(2): 289-300, 2023.

Purpose: This study aimed to test the hypothesis that a novel nutritional blend composed of tryptophan, glycine, magnesium, tart cherry powder, and L-theanine enhances subjective and objective measures of sleep during free living conditions.

Methods: In a randomized, repeated-measures crossover and double-blind deception design, participants (n = 9 males and 7 females, age = 24 ± 3 yr, body mass = 69.8 ± 11.6 kg, stature = 170.8 ± 9.1 cm) completed a 3-d familiarization period, followed by 3-d intervention and placebo trials. Subjective Pittsburgh Quality Sleep Index, Core Consensus Sleep Diary, and Karolinska Sleepiness Scale survey tools, alongside objective actigraphy measures of sleep, were assessed, with daily nutritional intake, activity, and light exposure standardized between trials. Participants provided daily urine samples for assessment of targeted and untargeted metabolomes.

Results: The intervention trial reduced sleep onset latency (−24 ± 25 min; P = 0.002), increased total sleep time (22 ± 32 min; P = 0.01), and increased sleep efficiency (2.4% ± 3.9%; P = 0.03), while also reducing morning sleepiness (P = 0.02). Throughout the study, 75% of participants remained blinded to sleep assessment as a primary outcome measure, with 56% subjectively indicating improved sleep during the intervention trial. Metabolomic analysis highlighted several significantly altered metabolomes related to sleep regulation between trials, inclusive of 6-sulfatoxymelatonin, D-serine, and L-glutamic acid.

Conclusions: Data demonstrate that using the proposed blend of novel nutritional ingredients during free living conditions reduced sleep onset latency, increased total sleep duration, and increased sleep efficiency, leading to reduced perceptions of morning sleepiness. These effects may be mediated by the up-regulation of key metabolites involved in the neurophysiological modulation of the sleep/wake cycle.

Sports Science Research Update

Pajamas, polysomnography and professional athletes: The role of sleep tracking technology in sport. Sports, 11(14): 1-17, 2023.

Abstract - Technological advances in sleep monitoring have seen an explosion of devices used to gather important sleep metrics. These devices range from instrumented 'smart pajamas' through to at-home polysomnography devices. Alongside these developments in sleep technologies, there have been concomitant increases in sleep monitoring in athletic populations, both in the research and in practical settings. The increase in sleep monitoring in sport is likely due to the increased knowledge of the importance of sleep in the recovery process and performance of an athlete, as well as the well-reported challenges that athletes can face with their sleep. This narrative review will discuss: (1) the importance of sleep to athletes; (2) the various wearable tools and technologies being used to monitor sleep in the sport setting; (3) the role that sleep tracking devices may play in gathering information about sleep; (4) the reliability and validity of sleep tracking devices; (5) the limitations and cautions associated with sleep trackers; and, (6) the use of sleep trackers to guide behavior change in athletes. We also provide some practical recommendations for practitioners working with athletes to ensure that the selection of such devices and technology will meet the goals and requirements of the athlete.

Sports Science Research Update

Regular-season injury rates in the National Football League after an attenuated preseason due to COVID-19. The Orthopaedic Journal of Sports Medicine, 10(11): 1-4, 2022.

Background: The preseason is a crucial time period for professional athletes to prepare for the upcoming season. However, due to the coronavirus 2019 (COVID-19) pandemic, the 2020 National Football League (NFL) preseason was canceled. As the regular season progressed, an increased number of injuries became apparent.

Purpose/Hypothesis: The purpose of this study was to compare injury rates between the 2019 and 2020 NFL regular seasons after the canceled 2020 preseason. It was hypothesized that injury rates in the 2020 season would be the same or higher compared with 2019.

Study Design: Descriptive epidemiology study.

Methods: A public database of all 32 NFL teams was used to examine the movement of all players to injured reserve status as well as every player who missed a regular-season game due to injury for both the 2019 and the 2020 NFL regular seasons. Data collection consisted of tallying the total number of reported musculoskeletal injuries for each team. Comparisons of injury rates between the 2 seasons were made using a Wilcoxon signed-rank statistical analysis.

Results: The total number of musculoskeletal injuries was 632 for the 2019 regular season and 765 for the 2020 regular season (P = .0324). Compared with 2019, statistically significant increases in injury rates in 2020 were seen for calf or Achilles tendon strain (17 vs 39; P = .009), groin injury (25 vs 46; P = .018), and anterior cruciate ligament tear (14 vs 29; P = .016). There were several other increased trends in soft tissue injury rates in the 2020 season (eg, hamstring injury [87 vs 114; P = .085]); however, these did not reach statistical significance.

Conclusion: This study showed a significant increase in total injuries during the 2020 COVID-19-affected NFL regular season compared with 2019. Several major soft tissue injuries reached statistical significance in their increase from 2019 to 2020 after the cancellation of the 2020 preseason due to COVID-19. These findings suggest that organized team preseason training and conditioning could have an effect on the prevention of sport-related injury at the highest level.

Sports Science Research Update

Relationships between repeated sprint ability, countermovement jump and thermography in elite football players. Sensors, 23(632): 1-9, 2023.

Abstract - Football is a very demanding sport which requires players to exert maximum effort, producing fatigue and eventually injuries. Thermography can be used to detect fatigue and prevent its consequences through thermal asymmetries in the bilateral body areas; however, its adequacy for elite footballers has not been widely studied. Therefore, the objective of the present investigation was to determine the suitability of thermography to detect fatigue in male football players. For this reason, twenty participants were gathered into a pair of subgroups (low [0.05) were found in any of the RSA test variables between low and high thermal asymmetry groups for thighs and calves. On the other hand, the low thermal asymmetry hamstring group reported a smaller percentage difference in sprints for the first sprint (%Diff) and a larger percentage difference in sprints two and three with respect to the best sprint (%Best). For CMJs, the low thermal asymmetry hamstring group reported significantly higher values post-RSA test, indicating better performance. Accordingly, thermography can provide information about performance in CMJ and RSA tests through hamstring asymmetries over 0.2 ◦C. Meanwhile, larger asymmetries than 0.2 ◦C in calves and thighs do not seem to be related to performance in these tests; therefore, coaches should consider if it is optimal to align players with high hamstring asymmetries.

Sports Science Research Update

Scheduling concurrent training 48 versus 72 hours after simulated match play: Effects on neuromuscular function and fatigue. Medicine and Science in Sports and Exercise, 55(2): 301-310, 2023.

Introduction: Scheduling concurrent training (CT) during the in-season microcycle in field-based team sport is driven by prematch and postmatch recovery. This study examined the neuromuscular function, fatigue, and soreness responses to CT administered 48 h (match day (MD) + 2) versus 72 h (MD + 3) after match.

Methods: Ten male recreational-level team sport athletes were monitored daily during two 5-d microcycles, which began with a simulated match (Soccer-specific Aerobic Field Test (SAFT90)) and CT performed either 48 or 72 h after match. Maximal voluntary force, quadriceps maximum EMG, voluntary activation, muscle contractile function (evoked twitch responses), muscle soreness, and fatigue were assessed immediately before and after the SAFT90, and every 24 up to 96 h after match. Outcome measures were also assessed immediately after CT. The CT consisted of an intermittent sprint protocol and a lower limb resistance training session separated by 1 h.

Results: Immediately after the SAFT90 in both conditions, maximal voluntary force was below baseline (mean change (Δ), −14.6% ± 10.0%; P = 0.03), recovering 48 h post. Quadriceps contractile function (Δ, −31.5% ± 11.4%; P = 0.003) and voluntary activation (Δ, −8.9 ± 6.2%; P = 0.003) were also hampered after the SAFT90, recovering 24 h post in both conditions. In addition, the SAFT90 elicited elevated levels of fatigue and muscle soreness that recovered 24 h after the SAFT90 before increasing at 72 and 96 h post in the MD + 2 and MD + 3 conditions, respectively.

Conclusions: Recovery of fatigue was only observed at the end of the microcycle when CT was prescribed on MD + 2. Therefore, CT scheduled early (MD + 2) in the microcycle might avoid compromising forthcoming match preparation.

Sports Science Research Update

The influence of injury history on countermovement jump performance and movement strategy in professional soccer players: Implications for profiling and rehabilitation Foci. Journal of Sport Rehabilitation, 30: 768-773, 2021.

Context: Professional soccer players who have sustained a lower limb injury are up to 3× more likely to suffer a reinjury, often of increased severity. Previous injury has been shown to induce compensatory strategies during neuromuscular screening tests, which might mask deficits and lead to misinterpretation of readiness to play based on task outcome measures.

Objective: To investigate the influence of previous injury in professional soccer players on countermovement jump (CMJ) performance and movement strategy.

Design: Cross-sectional.

Setting: Professional soccer club competing in the English Championship (tier 2).

Patients (or Other Participants): Outfield players with a minimum of 6 years as a professional.

Intervention(s): Players were categorized as previously injured (n = 10) or not injured (n = 10). All players completed double- and single-leg CMJ trials.

Main Outcome Measures: CMJ performance was quantified as jump height and flight time:contraction time ratio. CMJ movement strategy was quantified as force–time history, differentiating eccentric and concentric phases and CMJ depth.

Results: Double-leg CMJ was not sensitive to previous injury in performance or movement strategy. In contrast, single-leg CMJ performance was impaired in players with previous injury, who generated significantly lower eccentric and concentric peak force and rate of force development, and a deeper countermovement. Impaired single-leg CMJ performance was also evident in the nonaffected limb of previously injured players, suggesting cross-contamination. Hierarchical ordering revealed that the eccentric phase of the CMJ contributed little to performance in previously injured players. In noninjured players, the eccentric rate of force development and concentric peak force were able to account for up to 89% of the variation in CMJ performance.

Conclusions: Single-leg CMJ is advocated for player profiling, being more sensitive to previous injury, and negating the opportunity for interlimb compensation strategies. Movement strategy deficits in previously injured players suggest rehabilitation foci specific to eccentric force development.

Sports Science Research Update

The mechanism of anterior cruciate ligament injuries in the National Football League: A systematic video review. Cureus 15(1): e34291: 1-8, 2023.

Introduction: Anterior cruciate ligament (ACL) injuries are common among American football athletes, although few studies have performed video analyses of ACL injuries to better understand the injury mechanism. This work aims to characterize the mechanism of ACL injury during professional football competitions using video analysis. We hypothesize that football-specific injury trends will emerge, including high rates of contact injuries and associations with shallow knee and hip flexion angles (0°-30°).

Methods: Videos of professional football players suffering ACL injuries from 2007 to 2016 were analyzed. Injured players were identified using the injured reserve (IR) lists of the National Football League (NFL), and videos were discovered via a systematic Google search. Descriptive statistics and frequency analyses were performed on all variables using the Statistical Package for the Social Sciences (SPSS) version 23.0 (IBM SPSS Statistics, Armonk, NY, USA).

Results: Of the 429 ACL injuries identified, 53 (12%) videos were available. Deceleration was the most common injury maneuver, present in 32 (60%) athletes. Thirty-one (58%) players suffered contact injuries. Twenty-eight (53%) injuries demonstrated valgus collapse of the knee, and 26 (49%) had neutral knee rotation. Defensive backs (26%) and wide receivers (23%) were the most frequently injured positions.

Conclusion: Overall, we found that most ACL injuries had preceding contact, deceleration, shallow hip and knee flexion, and heel strike, and subsequent valgus collapse and neutral knee rotation. This understanding of American football-specific ACL tear mechanisms could help direct the focus of future injury prevention training modalities.

Sports Science Research Update

The use of biologics in NFL athletes: An expert consensus of NFL team physicians. The Orthopaedic Journal of Sports Medicine, 11(2): 1-15, 2023.

Background: There is a lack of published information outlining the use of biologics in National Football League (NFL) athletes and limited data to guide biologic treatment strategies. Purpose: To develop a consensus on the use of biologics among NFL team physicians. Study Design: Consensus statement.

Methods: A working group of 6 experts convened a consensus process involving NFL team physicians using validated Delphi methodology. Physicians from 32 NFL teams as well as NFL London were invited to take part. This iterative process was used to define statements on the use of biologics in NFL athletes. A recent scoping review exploring biologics in professional athletes was used to inform the first of 3 rounds of surveys, with statements considered under 7 headings: biologics in general, challenges of treating NFL athletes, terminology/nomenclature, autologous blood products, cell-based therapies, guidance for NFL team physicians, and biologic research in the NFL. In addition to rating agreement, experts were encouraged to propose further items or modifications. Predefined criteria were used to refine item lists after each survey. For a consensus within the final round, defined a priori, items were included in the final information set if a minimum of 75% of respondents agreed and fewer than 10% disagreed.

Results: Physicians from 26 NFL teams and NFL London responded to the initial invitation to participate in the Delphi process; 88.9% of participating team physicians completed the round 1 survey, with response rates of 87.5% in round 2 and 95.2% in round 3. After 3 rounds, 47 statements reached a consensus. A consensus was achieved that platelet-rich plasma has a positive impact on patellar tendinopathy and on symptoms in early osteoarthritis but not for other indications. NFL team physicians agreed that while cell therapies have the potential to improve symptoms, the misrepresentation of uncharacterized preparations as “stem cells” has contributed to the widespread use of unproven therapies.

Conclusion: This study established an expert consensus on 47 statements relating to the use of biologics in NFL athletes. In addition to providing clinical guidance for the use of biologics in NFL athletes, this study identified key areas for future focus including the development of athlete education materials.

Sports Science Research Update

Two or four weeks acute:chronic workload ratio is more useful to prevent injuries in soccer? Applied Sciences, 13(495): 1-8, 2023.

Abstract - This study was conducted to determine if the acute: chronic workload ratio (ACWR) is related to the incidence of non-contact injuries. The purpose is to compare the external load of injured and non-injured soccer players with the same characteristics, such as position and age. The present analysis considers both the four and the two weeks preceding an injury. Physical characteristics were recorded and analyzed through global positioning systems (GPS) evaluation over one season of 24 competitive microcycles, 144 training sessions, and 32 matches in a total of 35 professional soccer players from the Greek Super League 1 and Super League 2. The loads calculated were total distance (TD), 15–20 km/h, 20–25 km/h, 25–30 km/h, accelerations (ACC) > 2.5 m/s^2 , and decelerations (DEC) > 2.5 m/s^2). Nine injured athletes exceeded the critical threshold of an ACWR > 1.3 several times compared with non-injured athletes that did not reach this level. The present study showed that ACWR is related to a subsequent occurrence of injury but that the threshold of an ACWR can vary. This seems to be mainly influenced by assessing the load of the last two weeks compared with that of the four weeks before the injury.

Sports Science Research Update

Weekly external load performance effects on sports injuries of male professional football players. International Journal of Environmental Research and Public Health, 20(1121): 1-12, 2023.

Abstract - One of the most challenging issues professional football players face throughout their careers is injuries. Those injuries often result from suboptimal training programs that were not designed according to the players' individual needs. This prospective study aimed to examine in detail the effects of sports injuries on professional football players' weekly external load performances. Thirty-three male professional football players were monitored using 10-Hz Global Positioning System (GPS) units (Apex pro series, StatSports) during an entire season. The variables considered in the analysis were total distance (TD), high-speed running (HSR), accelerations (ACC), and decelerations (DEC). The comparisons were made between the four-week block before injury (–4T), four-week block after return (+4T), and players' season averages (S). Players displayed significantly higher values of TD, HSR, ACC, and DEC in the –4T, compared to the other two moments (+4T and S). Furthermore, the comparison between the +4T and S showed no significant variations in the GPS metrics. It was shown that a significant increase in players' weekly external load performance over a four–week period may have a negative effect on the occurrence of injuries from a professional football standpoint. Future research should consider the effects of injury severity on players' external load variations.